Heat Transfer · Worksheet 3

Conduction through plane walls · Sample document bundled with Nib-E

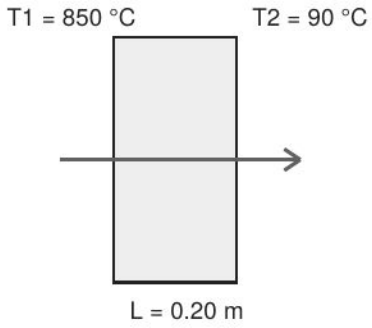
Name

Matriculation no.

☐ I solved these problems on my own

Problem 1

A furnace wall is 0.20 m thick with thermal conductivity k = 1.2 W/(m·K).
The inner surface is held at 850 °C and the outer surface at 90 °C.
Find the heat flux q" through the wall using Fourier's law: q" = k (T1 − T2) / L.



Problem 2

Add a 0.05 m insulation layer with k = 0.08 W/(m·K) on the outside.
Keeping the same surface temperatures, by what factor does the heat loss drop?

Tip: pick the pen and write your working on the lines. Fill the fields above with the hand tool.